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Object: Submission of the article “Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization”.

Dear Editor,

We wish to submit a new research paper entitled “Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization” for consideration by the *IEEE Transactions on Communications*. We confirm that this work is original and has not been published elsewhere nor is it currently under consideration for publication elsewhere.

This manuscript investigates advanced methods for indoor optical wireless positioning. While state-of-the-art approaches typically rely on dense multi-LED infrastructures or complex receiver arrays to achieve high accuracy, these requirements impose significant hardware costs and deployment constraints. To address this limitation, we propose a novel positioning architecture that achieves three-dimensional localization using a minimal hardware configuration: a single beam-steered LED and a single photodiode, without requiring receiver rotation or complex receiver arrays. We develop a comprehensive theoretical framework for this system, deriving the composite Cramér-Rao lower bound and Position Error Bound to characterize the fundamental limits of the proposed joint observation model. Furthermore, we address the system design by casting the steering-pattern selection as a genetic algorithm optimization problem. Finally, we derive and compare different estimation strategies, including a constrained nonlinear estimator and two closed-form direction estimators. The aim of this analysis is to demonstrate that centimeter-level accuracy is achievable with reduced infrastructure complexity by leveraging transmitter beam-steering capabilities.

Given the increasing relevance of integrated communication and sensing (ISAC) in future 6G networks, we believe this paper is of strong interest to readers in the areas of optical wireless communications, LiFi, ISAC, and indoor localization, placing this work squarely within the scope of the journal.

This manuscript is the work of Kevin Acuna-Condori (corresponding author), Bastien Béchadergue, Hongyu Guan, and Luc Chassagne. Please address all correspondence concerning this paper to me at kevin-jose.acuna-condori@uvsq.fr.

Thank you for your consideration of this manuscript.

Sincerely,



Mr. Kevin Acuna-Condori